Cassie Gallogly

Berkeley, CA | 310-972-9826 | cassiegallogly21@gmail.com

TECHNICAL SKILLS: C/C++, Python, Javascript, Verilog, Matlab, RISC-V, MuJoCo, CoppeliaSim, LaTeX, and OnShape.

CORE COMPETENCIES: Systems Integration & Reliability Testing, Sensor-Driven Feedback Control, Human-Centered Technology Development, Technical Mentorship

ENGINEERING PROJECTS

TACTILE MANIPULATION RESEARCH LAB June 2025 - Present

- Designed and fabricated piezoresistive soft tactile sensors using carbon black/Ecoflex composites with diamond-lattice internal channel geometry for embedded-wiring-free deformation and force detection.
- Developed independent research plan and experimental protocol; validating sensing accuracy, repeatability, and dynamic force response toward a peer-reviewed publication.

PROSTHETIC FOOT SUSPENSION SYSTEM Fall 2025 - June 2026

- Designed an adaptive pneumatic suspension system for transtibial prostheses, improving user stability across variable terrain.
- Developed microcontroller-based sensor feedback for real-time damping adjustment and gait performance optimization.
- Validated a low-cost, universally compatible solution through iterative mechanical testing and performance analysis.

AUTONOMOUS MECHATRONICS PROJECT Spring 2025

- Designed and built a fully autonomous bartender robot integrating fluid dispensing actuators and embedded control systems for real-time drink selection and delivery.
- Implemented sensor fusion and closed-loop feedback to achieve repeatable, precise pour accuracy across varying conditions.
- Collaborated in a multidisciplinary team to deliver a functional end-to-end system with documented, maintainable firmware and mechanical design.

EDUCATION

UNIVERSITY OF CALIFORNIA - BERKELEY Berkeley, CA
Masters of Engineering, Mechanical Engineering Present - May 2027
Concentration in Control of Robotic and Autonomous Systems

UNIVERSITY OF CALIFORNIA - SANTA CRUZ Santa Cruz, CA
Bachelor of Science, Robotic Engineering; Minor in Electrical Engineering Sept 2022 - June 2026
GPA: 3.7, Dean's Honor List

MIRA COSTA HIGH SCHOOL Manhattan Beach, CA
GPA: 4.2, Dean's List & Honors Society Aug 2018 - June 2022

WORK EXPERIENCE

Baskin Engineering at UCSC Santa Cruz, CA
Undergraduate Researcher and Group TA June 2025 – January 2026

- Conducted robotics research and system testing; mentored students in simulation and embedded systems coursework.

SHARKS ICE RINK San Jose, CA
Professional Figure Skating Coach Nov 2023 – Present

- Designed and delivered private lessons in technique and performance; coordinated training programs across a team of instructors.

AWARDS

ACADEMIC: Dean's Honors List 2022 - 2026, National Honors Society 2018-2022
ATHLETIC: US Figure Skating National Showcase Medalist 2022, US Figure Silver Skating Awards 2022, ISI Worlds 2024